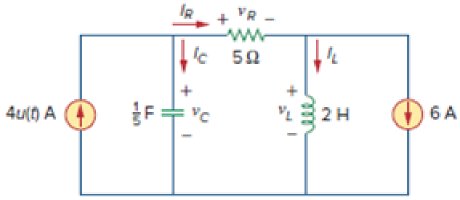
Chapter 8, Problem 2PP

Problem

For the circuit in Fig. 8.7, find: (a) $i_L(0+)$, $v_C(0+)$, $v_R(0+)$, (b) $di_L(0+)/dt$, $dv_C(0+)/dt$, $dv_R(0+)/dt$, (c) $i_L(\infty)$, $v_C(\infty)$, $v_R(\infty)$.

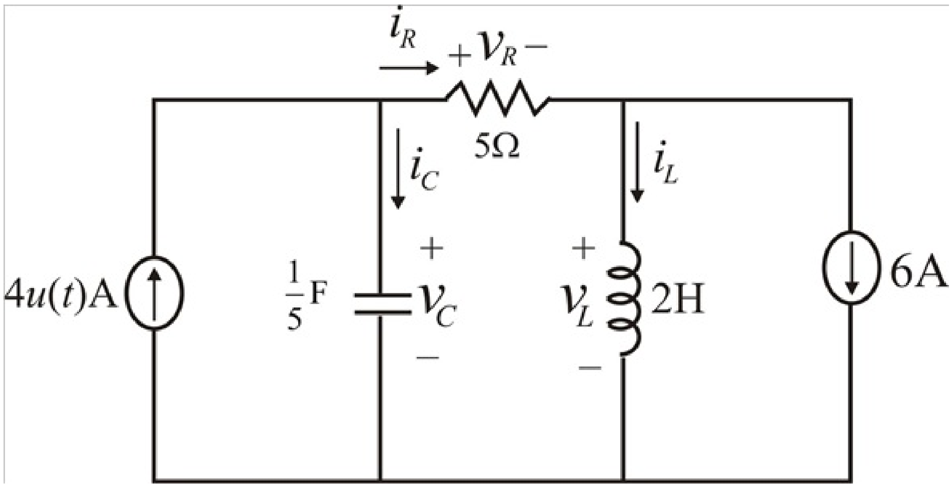
Figure 8.7:



Step-by-step solution

Step 1 of 13

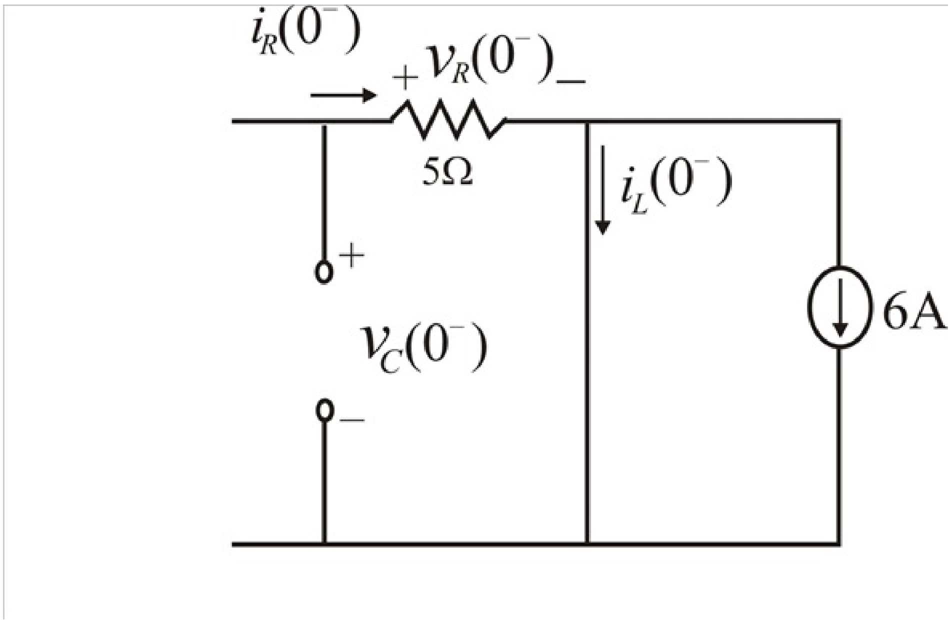
The given circuit is



Step 2 of 13

(a) For $t>0$, $4u(t)=0$

At $t=0^-$, since the circuit has reached steady state, the inductor can be replaced by a short circuit, while the capacitor is replaced by an open circuit. Therefore the circuit is as shown.



Step 3 of 13

From above circuit, we obtain

$i_L(0^-)=-6\text{A}$

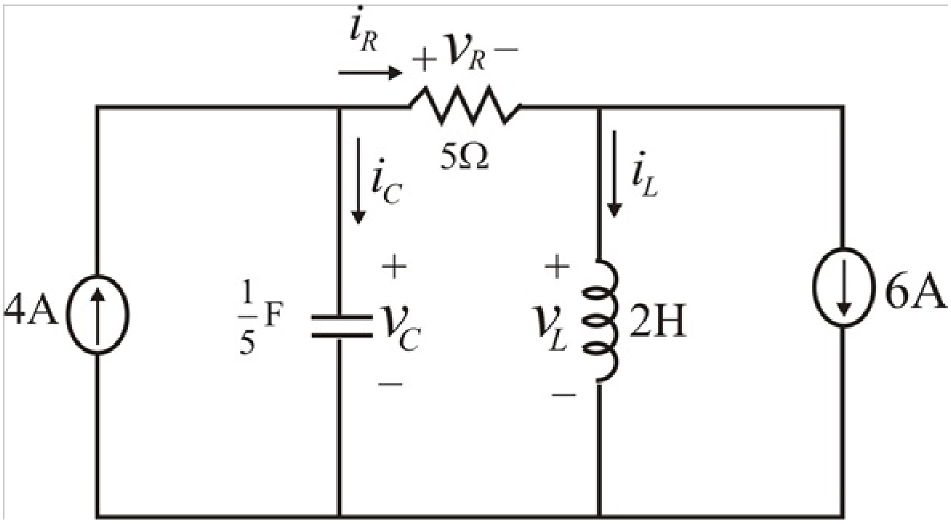
$i_R(0^-)=0\text{A}$

$v_R(0^-)=0\text{V}$

$v_C(0^-)=0\text{V}$

Step 4 of 13

For $t>0$, $4u(t)=4$, so that the circuit is now as shown below



Step 5 of 13

Since the inductor current and capacitor voltage cannot change abruptly,

$$i_L(0^+) = i_L(0^-) = -6\text{A}$$

$$v_C(0^+) = v_C(0^-) = 0\text{V}$$

$$\therefore \boxed{i_L(0^+) = -6\ \text{A}}$$

$$\boxed{v_C(0^+) = 0\ \text{V}}$$

Step 6 of 13

Since

$v_C(0^+) = 0\text{V}$ i.e. a short circuit across 4A current source, so there is no current through the resistor. Therefore,

$$i_R(0^+) = 0\text{A}$$

$$i_C(0^+) = 4\text{A}$$

$$v_R(0^+) = 5i_R(0^+)$$

$$v_R(0^+) = 5 \times 0$$

$$\therefore \boxed{v_R(0^+) = 0\ \text{V}}$$

Step 7 of 13

(b) The voltage across the inductor is

$$v_L = L\frac{di_L}{dt}$$

$$\frac{di_L(0^+)}{dt} = \frac{v_L(0^+)}{L}$$

Step 8 of 13

Applying KVL to the center mesh

$$-v_C(0^+) + v_R(0^+) + v_L(0^+) = 0$$

$$v_L(0^+) = v_C(0^+) - v_R(0^+)$$

$$v_L(0^+) = 0 - 0$$

$$v_L(0^+) = 0\text{V}$$

$$\therefore \boxed{\frac{di_L(0^+)}{dt} = 0\ \text{A/s}}$$

Step 9 of 13

The current through the capacitor is

$$i_C(0^+) = C\frac{dv_C(0^+)}{dt}$$

$$\frac{dv_C(0^+)}{dt} = \frac{i_C(0^+)}{C}$$

We have obtained,

$$i_C(0^+) = 4\text{A}$$

$$\frac{dv_C(0^+)}{dt} = \frac{4}{\left(\frac{1}{5}\right)}$$

$$\therefore \boxed{\frac{dv_C(0^+)}{dt} = 20 \text{ V/s}}$$

Step 10 of 13

Applying KCL at the top right node,

$$-i_R + i_L + 6 = 0$$

Taking the derivative of each term on both sides and setting $t = 0^+$ gives

$$-\frac{di_R(0^+)}{dt} + \frac{di_L(0^+)}{dt} + 0 = 0$$

$$-\frac{di_R(0^+)}{dt} + \frac{di_L(0^+)}{dt} = 0$$

$$\frac{di_R(0^+)}{dt} = \frac{di_L(0^+)}{dt}$$

$$\frac{di_R(0^+)}{dt} = 0 \text{ A/s}$$

Step 11 of 13

Now we have,

$$v_R = 5i_R$$

Again taking derivative of each term and setting $t = 0^+$ gives

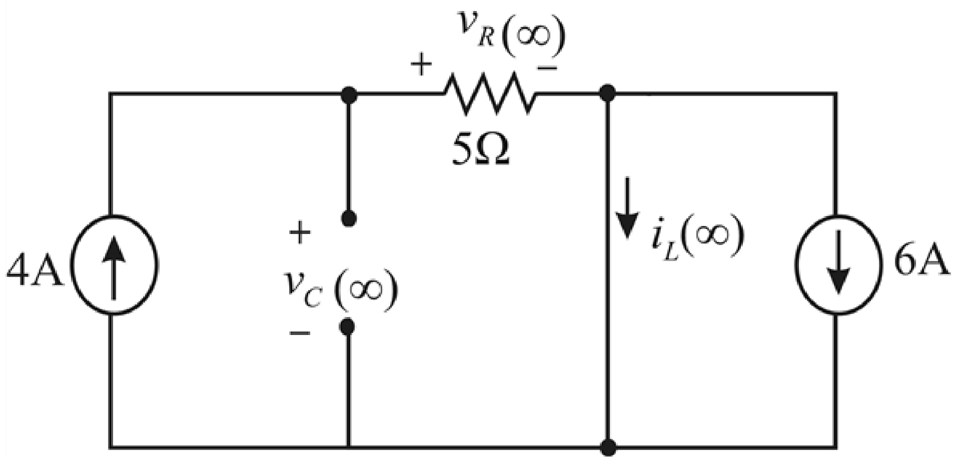
$$\frac{dv_R(0^+)}{dt} = 5\frac{di_R(0^+)}{dt}$$

$$\frac{dv_R(0^+)}{dt} = 5 \times 0$$

$$\therefore \boxed{\frac{dv_R(0^+)}{dt} = 0 \text{ V/s}}$$

Step 12 of 13

(c) As $t \to \infty$, the circuit reaches steady state. Therefore the inductor is replaced with a short circuit and capacitor is replaced with an open circuit. The equivalent circuit is as shown.



Step 13 of 13

Writing KCL at the top right node

$$-4 + i_L(\infty) + 6 = 0$$

$$\therefore \boxed{i_L(\infty) = -2\ \text{A}}$$

The current 4A flows through the 5Ω resistor. So,

$$v_R(\infty) = 5 \times 4$$

$$\therefore \boxed{v_R(\infty) = 20\ \text{V}}$$

Due to the short circuit,

$$v_C(\infty) = v_R(\infty)$$

$$\therefore \boxed{v_C(\infty) = 20\ \text{V}}$$